

Arya Raut

608 982 3064 | aryaraut11@gmail.com | Madison, WI | linkedin.com/in/arya-raut

EDUCATION

University of Wisconsin-Madison

Master of Science, Industrial Engineering

May 2026

GPA: 3.75/4

University of Mumbai

Bachelor of Technology, Mechanical Engineering & Minors in Data Science

June 2024

GPA: 3.56/4

WORK EXPERIENCE

Sani-Matic

Sep 2025 – Dec 2025

Continuous Improvement Engineer Consultant

Sun Prairie, WI

- Redesigned a 48 step sheet metal fabrication workflow using PDCA and time studies, eliminated 9 non-value added steps to reduce cycle time by **37%** per part, saving **\$60K** annually.
- Led cross-functional automation through manufacturing system integration, implemented **APQP** to ensure quality reliability, reducing waste by **7%** and saving **240** annual labor hours.
- Drafted **SWI** (standardized work instructions), validation protocols, implemented **SPC** to reduce defect rates, and improved production reliability by **22%**.

Project Assistant - Grader

Aug 2025 – Dec 2025

Courses - Engineering Management for Continuous Improvement, Quality Engineering & Quality Management

- Evaluated graduate-level coursework assignments focused on Lean manufacturing, Kaizen, Six Sigma, Quality Control, **Minitab** assessments & process optimization applied to industrial projects.

Doosan Bobcat

Sep 2024 – Dec 2024

Continuous Improvement Consultant

Johnson Creek, WI

- Led cross-functional Kaizen team to analyze tire handling operations using 5S, PDCA cycle frameworks, and defined Control Plan, KPI for continuous monitoring and sustained improvement.
- Performed **RCA** (Root Cause Analysis) using spaghetti diagrams, time study analysis, and ergonomic assessments, identifying critical bottlenecks through Pareto Analysis, 8D, and 5 Whys techniques.
- Implemented Lean Six Sigma methodologies to develop a low-cost material handling solution for tire lifting, reducing worker injury risk, and improving process efficiency by **7.83%**.

Pragati Switchgears

Jun 2024 – Aug 2024

Industrial Engineering Intern

Mumbai, India

- Conducted **VSM** (Value Stream Mapping) and designed custom fixtures using SolidWorks, Fusion 360, and AutoCAD, streamlining the Centrifugal Switchgear process and increasing the production rate by **21.6%** per hour.
- Implemented circuit breaker production assembly line from scratch, with electromechanical component integration, testing, and validating **DFM** (Design for Manufacturability).
- Conducted **PFMEA** to identify failure modes in Terminal Block assembly and implemented corrective fixture redesigns using SolidWorks, improving line efficiency by **15.66%**.

Flora Super Prints

Jun 2023 – Sep 2023

Operations Analyst Intern

Mumbai, India

- Designed an ergonomic trolley for bottle transport between labeling, curing furnace, and packaging lines, reducing handling time by **18.93%** and improving worker safety by **35.38%**, based on **NASA-TLX** assessment.
- Optimized inventory flow using **Kanban** principles and restructured stack organization by order deadlines, saving **140** minutes per worker daily, and ensuring on-time delivery.
- Leveraged Excel's Open Solver to optimize labeling team schedules, mitigating labor absences and increasing **OEE** (Operator Equipment Efficiency) by **11%**.

SKILLS AND CERTIFICATIONS

• **Software & Programming Languages:** AutoCAD, Solidworks, SAP, Catia, SQL, Minitab, ANSYS, Tableau, MS Office, Power BI, Trello, C, C++, MATLAB, Python, SQL, R, R Shiny

• **Awards & Certificates** | ISO 9001:2015 & ISO 14001:2015 Auditor, Lean Six Sigma Green Belt, SAP

PROJECTS

DJS Skylark | SAE Aero Design Competition

- Led the Structure & Stability team, mentoring six members while driving aircraft structural design, assembly, and troubleshooting to improve stability and overall performance.
- Secured **1st** and **3rd** place in design reports and mission performance at the SAE Aero Design Competition, built a Python optimization script for tail moment, payload configuration, and planform analysis, reducing testing time by 30%.

Downtime Optimization & Cost Reduction in Production Systems

- Developed a **Mixed-Integer Linear Programming** (MILP) model using Python to optimize production scheduling and task sequencing across 52 process stations for vehicle assembly line, reducing idle time by **12%**.
- Improved throughput by **15%**, reduced annual cost by **\$0.7M**, and conducted sensitivity analysis on downtime & energy costs.